Edwin Chen

347-379-5481 | ec769@cornell.edu | linkedin.com/in/edwin-chen

Education

Cornell University, College of Engineering – *Ithaca, NY* **Jan 2026 – Expected Dec 2026**
Masters of Engineering in Electrical and Computer Engineering

Cornell University, College of Engineering – *Ithaca, NY* **Aug 2022 – Dec 2025**
Bachelors of Science in Electrical and Computer Engineering, Minor in Computer Science *GPA 3.9/4.0*

Relevant Courses: Memory Centric Computing, Computer Architecture, Digital VLSI, Analog Integrated Circuits, High Level Synthesis, Systems Programming, Microelectronics, Signals & Systems, Embedded Systems, Digital Logic & Computer Organization, Circuits, Data Science, Data Structures & Object Oriented Programming, Linear Algebra, Differential Equations

Research and Experience

NASA – *Control Electronics Intern – Greenbelt, Maryland* **June 2025 – Aug 2025**

- Improved robot arm navigation with Q-learning using ϵ-greedy and softmax policies, validated in obstacle environments.
- Tested autonomous obstacle avoidance by simulating robotic arm navigation with Q-learning with ROS, MoveIt, and RViz.
- Redesigned rover electronics to integrate a four-switch buck-boost converter on a Raspberry Pi 5, improving power flexibility.

Cornell Custom Silicon Systems – *Physical Design Engineer* **Aug 2025 – Present**

- Assisted physical design efforts for digital, analog, and RF tapeouts on TSMC 180nm process node.
- Created and managed RTL → GDS design flow, physical verification, padring and packaging design.

Space Systems Design Studio – *Undergraduate Researcher* **June 2024 – May 2025**

- Designed impedance matching circuit for compact PCB for ChipSat to enhance antenna signal integrity
- Validated power requirements by developing a circuit to simulate and measure consumption during a 3-hour balloon test flight.
- Calibrated and tuned antennas with a NanoVNA to ensure optimal return loss at LoRa and L1 GPS frequencies.

Cornell Neuro-AI Lab – *Undergraduate Researcher* **June 2024 – Aug 2024**

- Designed and implemented a data conversion module to standardize input/output by validating and converting between Python and NumPy datatypes, accounting for NumPy bit-widths to ensure compatibility, efficiency, and type-range enforcement.
- Engineered a touch sensor capable of detecting tactile input from rats on a Teensy 4.1 microprocessor.

Cornell Autonomous Bicycle – *Embedded Software Engineer* **Sep 2023 – Dec 2024**

- Integrated a GPS unit with the Arduino Vidor 4000 via the I2C protocol, and validated accurate date and time data.
- Coordinated with a team of 30 engineers across electrical, mechanical, and software teams to develop an autonomous bicycle.

ECE Undergraduate Teaching Assistant **Jan 2024 – Present**

- Teaching Assistant for ECE 4750: Computer Architecture (Fall '25), ECE 3150: Microelectronics (Spring '25), and ECE 2100: Intro to Circuit Analysis (Spring '24 and Fall '24)

Design Projects

6T SRAM Array Design **Feb 2025**

- Designed an 6T SRAM bitcell in Cadence Virtuoso (gpdk45 PDK) through schematic, layout, and extraction.
- Implemented column circuitry (write driver, sense amp, precharge, reset, and output latch) with optimized drive strength.
- Constructed a 32 × 32 SRAM array with custom decoder and control circuitry, achieving clean DRC and LVS.
- Analyzed post-layout timing and validated stable read/write operation under Monte Carlo simulation.

SRAM Compute-In-Memory **April 2025**

- Designed a 10T XNOR-SRAM bitcell in Cadence Virtuoso (asap7 PDK) and simulated XAC operations on Synopsys HSPICE.
- Validated analog compute-in-memory behavior by characterizing transfer curves, delay and power at different supply voltages.

8-bit Multiply and Accumulate Cell **May 2025**

- Designed 8-bit carry-save multiplier, 20-bit carry-select adder, and accumulator register in Cadence Virtuoso (gpdk45 PDK).
- Completed schematic, layout, and post-layout extraction of the full MAC cell with hierarchical sub-circuits.
- Verifying correct MAC accumulation over 10 cycles and measured delay and energy at schematic and extracted levels.

Multicore Pipelined Processor **Dec 2024**

- Designed a 5-stage full-bypass processor with two-way associative cache into a single core architecture.
- Integrated four cores into a multi-core processing system, analyzing parallel speedup and ensuring cache coherence.

Awards

- **NASA Proposal Writing:** Presented proposal 'V.A.A.C.E. Venus Aerobot for Atmosphere and Cloud Exploration," successfully securing $10,000 in funding.
- Cornell University Dean's List (All Semesters)

Technical Skills

Languages: Verilog, Python, C/C++, TCL, Java, Latex, Linux, Git
Technical: Cadence (Virtuoso, Innovus) , Synopsys (VCS, Design Compiler, PrimeTime, Hspice), FPGA (Altera and Xilinx), Embedded OS, Microcontrollers, Altium, PCB Design, LTSpice